

Thesis Structure Map

Recall-Oriented TDA for Wearable-Sensor Fall Detection

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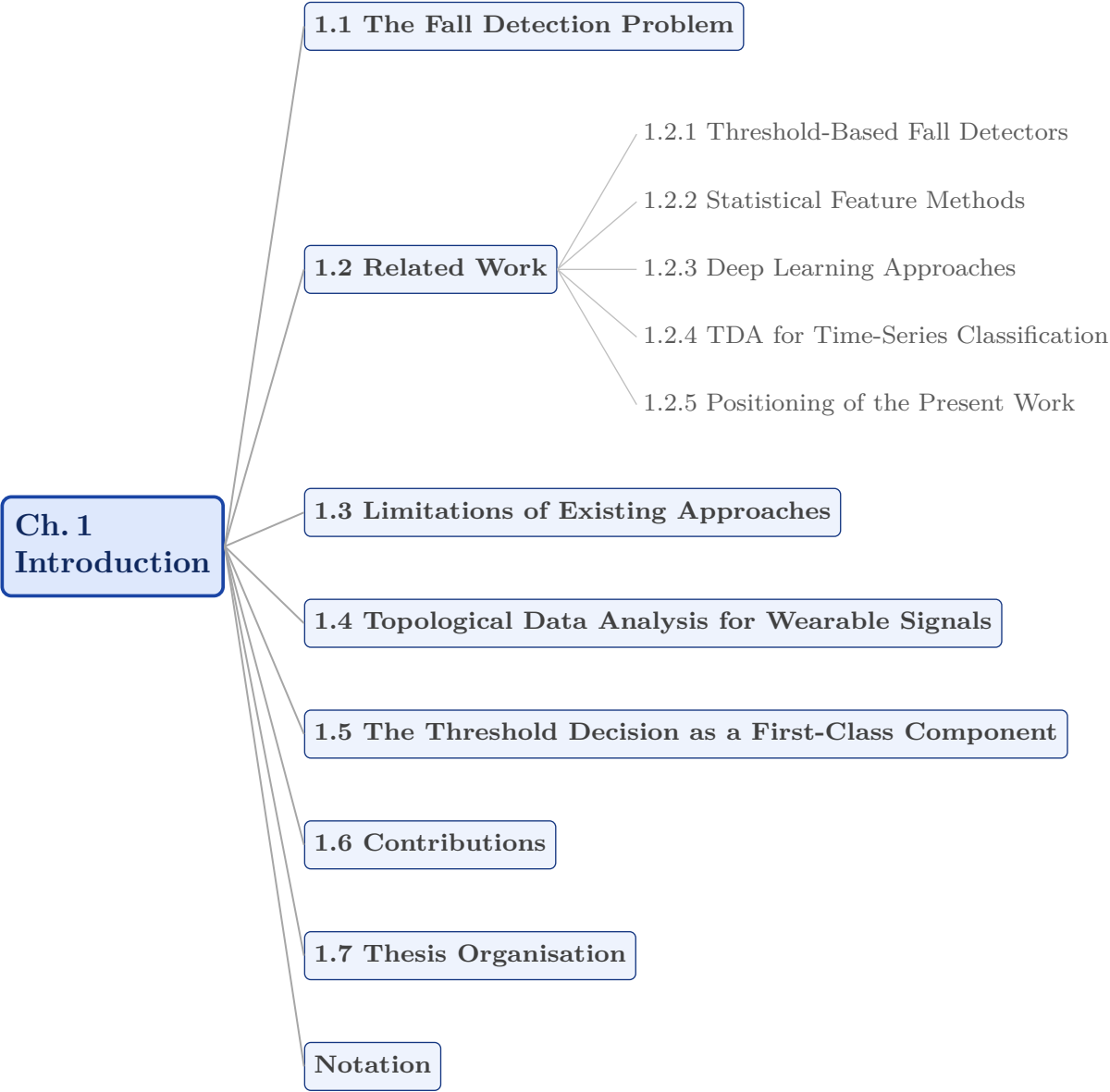
How to read this map.

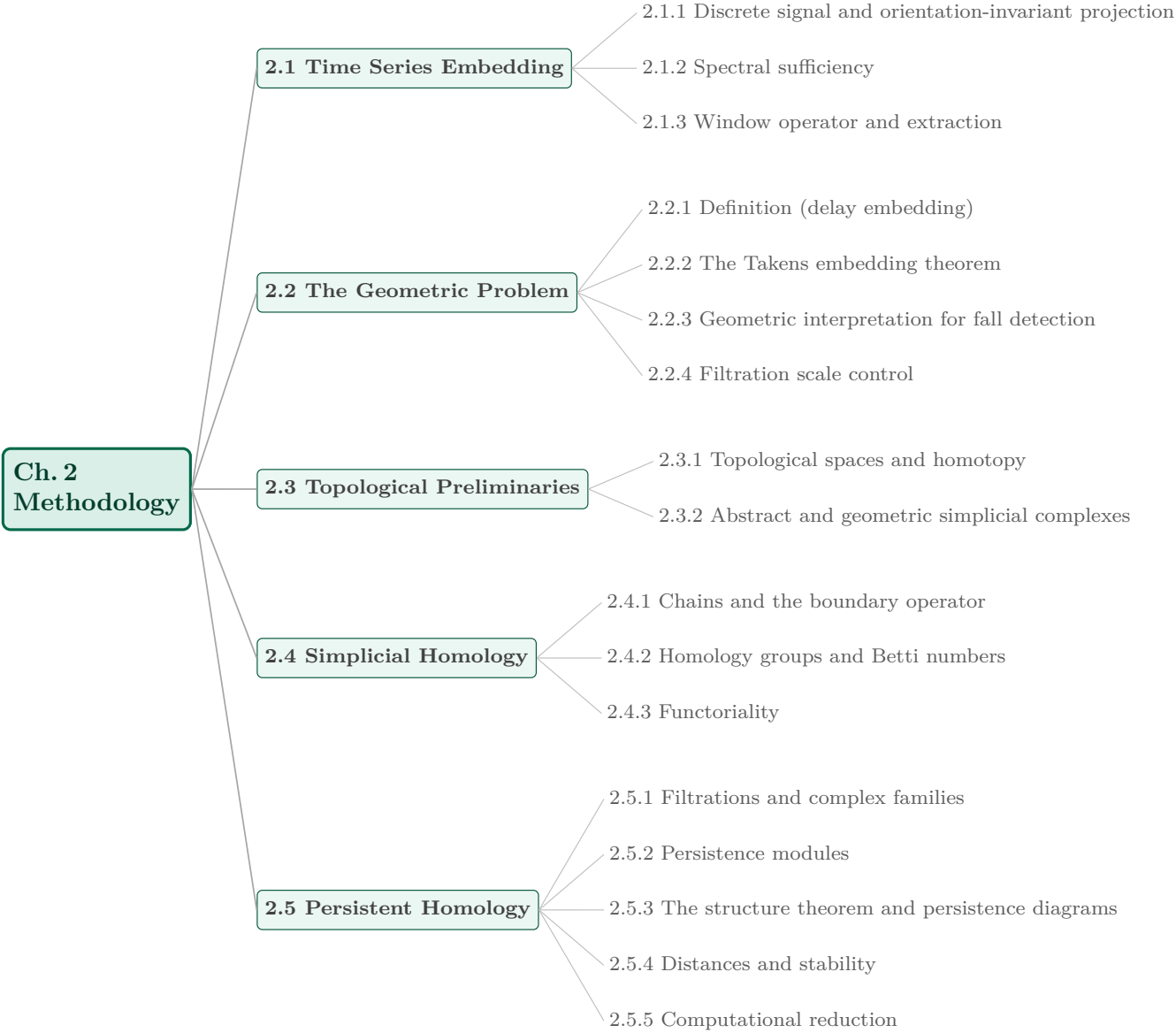
This document has two parts.

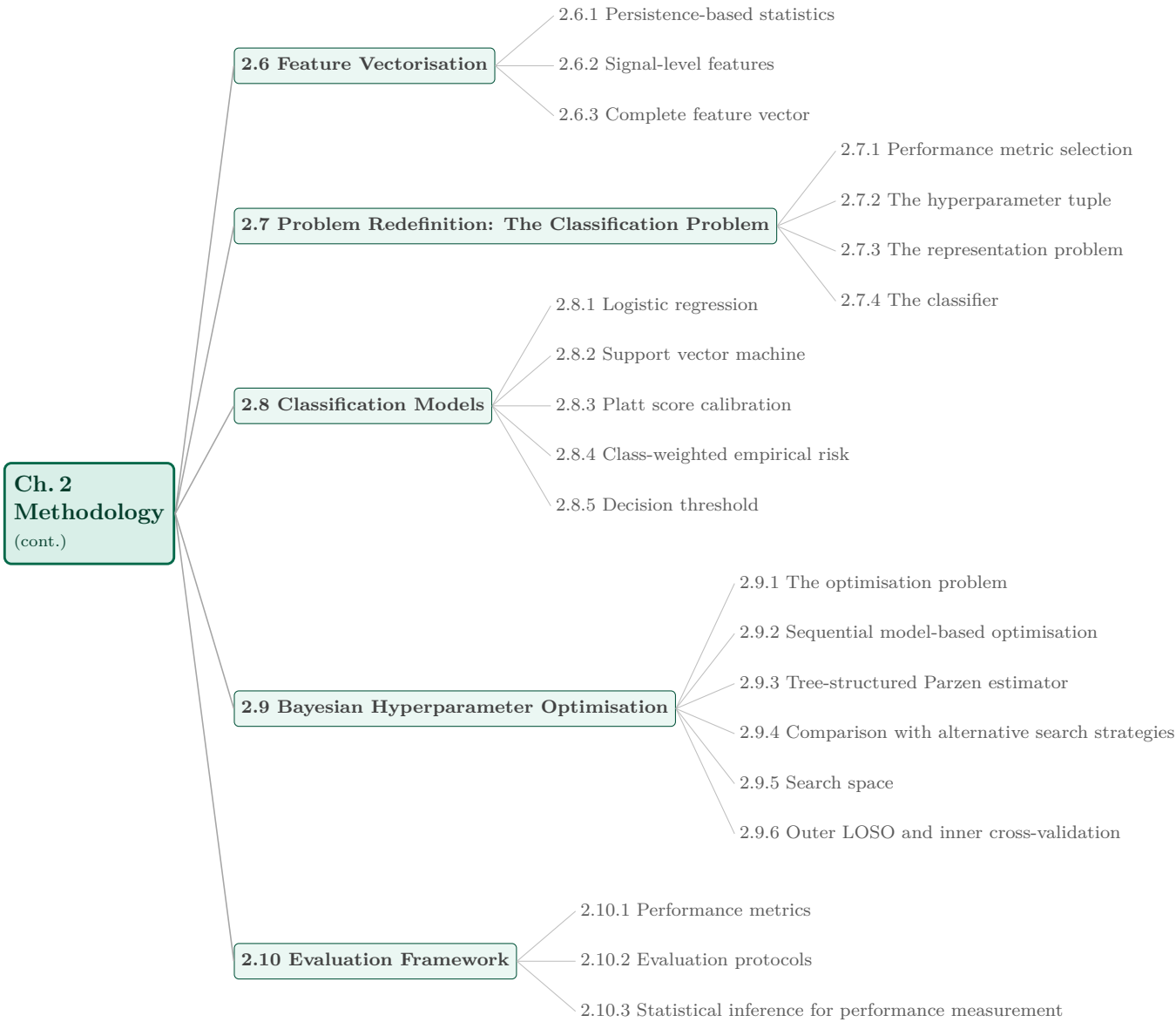
Part A (next 6 pages) gives one **section/subsection tree** per chapter: chapter title on the left, its sections one level in, its subsections one level further in. This is the literal table of contents, redrawn as a tree so the hierarchy is visible at a glance.

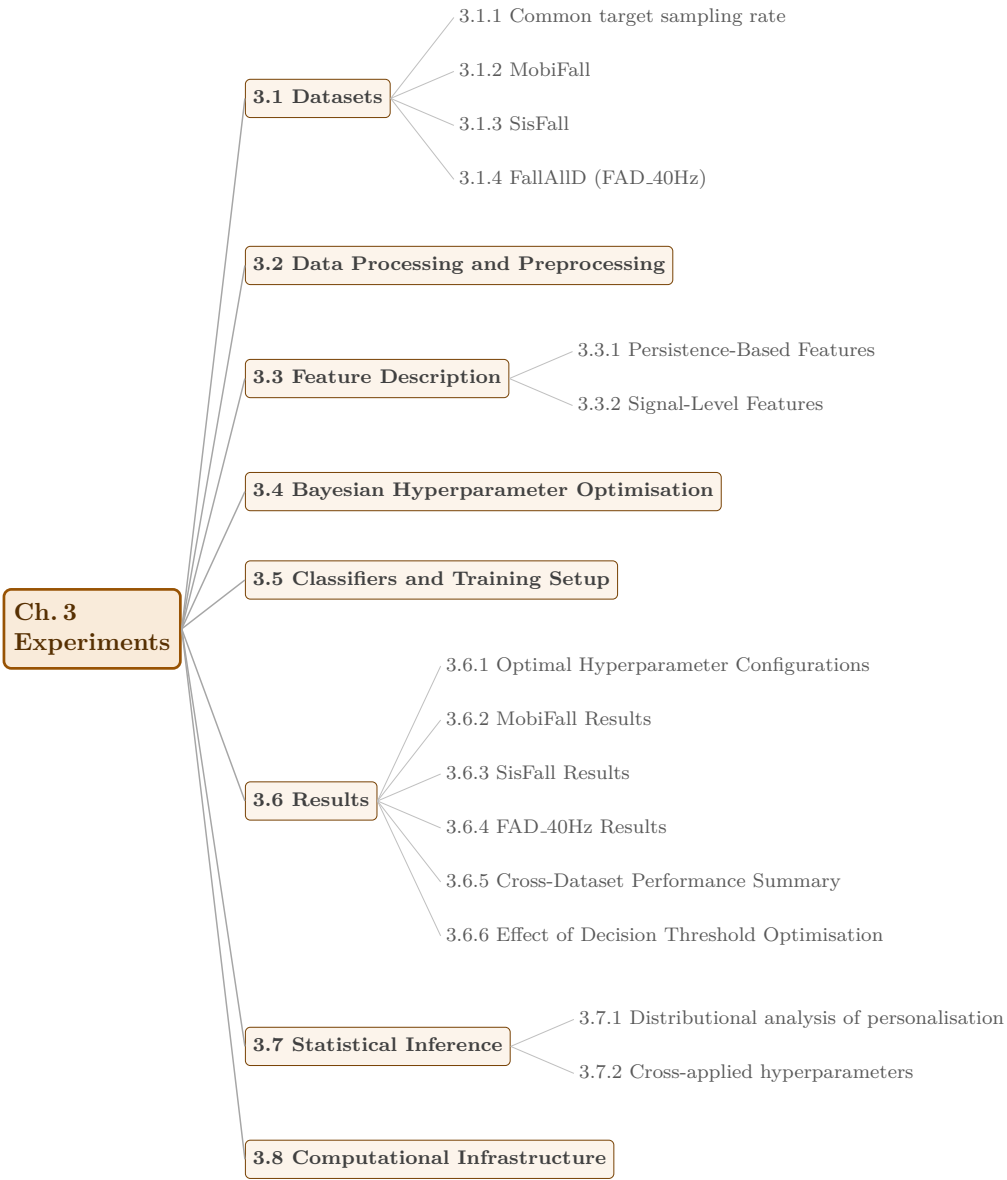
Part B (remaining pages) explains **how the six trees connect**: which chapter's output feeds the next chapter's input, why the Methodology chapter is ordered the way it is (simple concepts before complex ones, in the order they are actually used in the pipeline), and which abstract object defined in Methodology is later instantiated with concrete numbers in Experiments / discussed in Results.

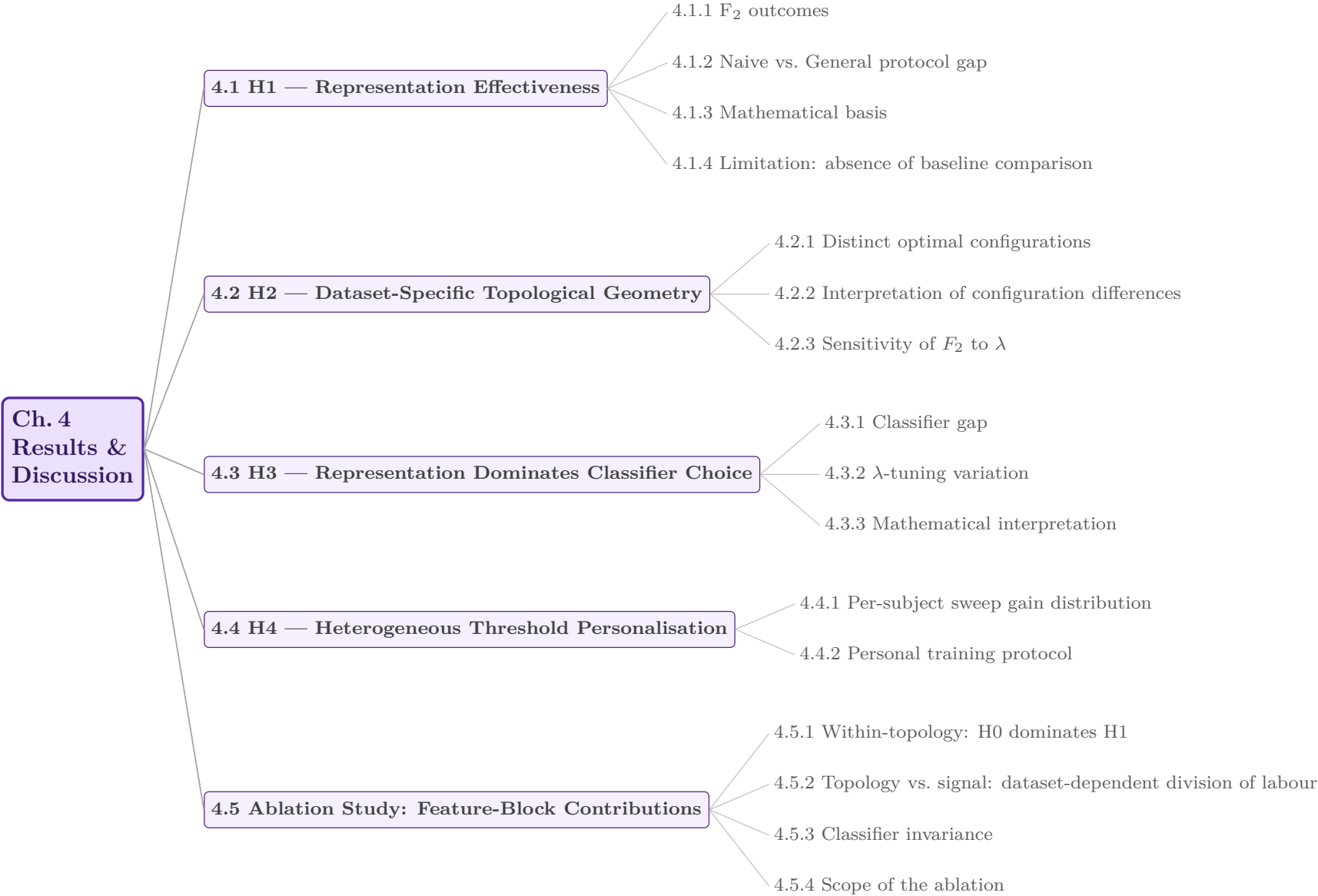
Colour key:  Ch.1 Introduction  Ch.2 Methodology  Ch.3 Experiments  Ch.4 Results & Discussion  Ch.5 Conclusion

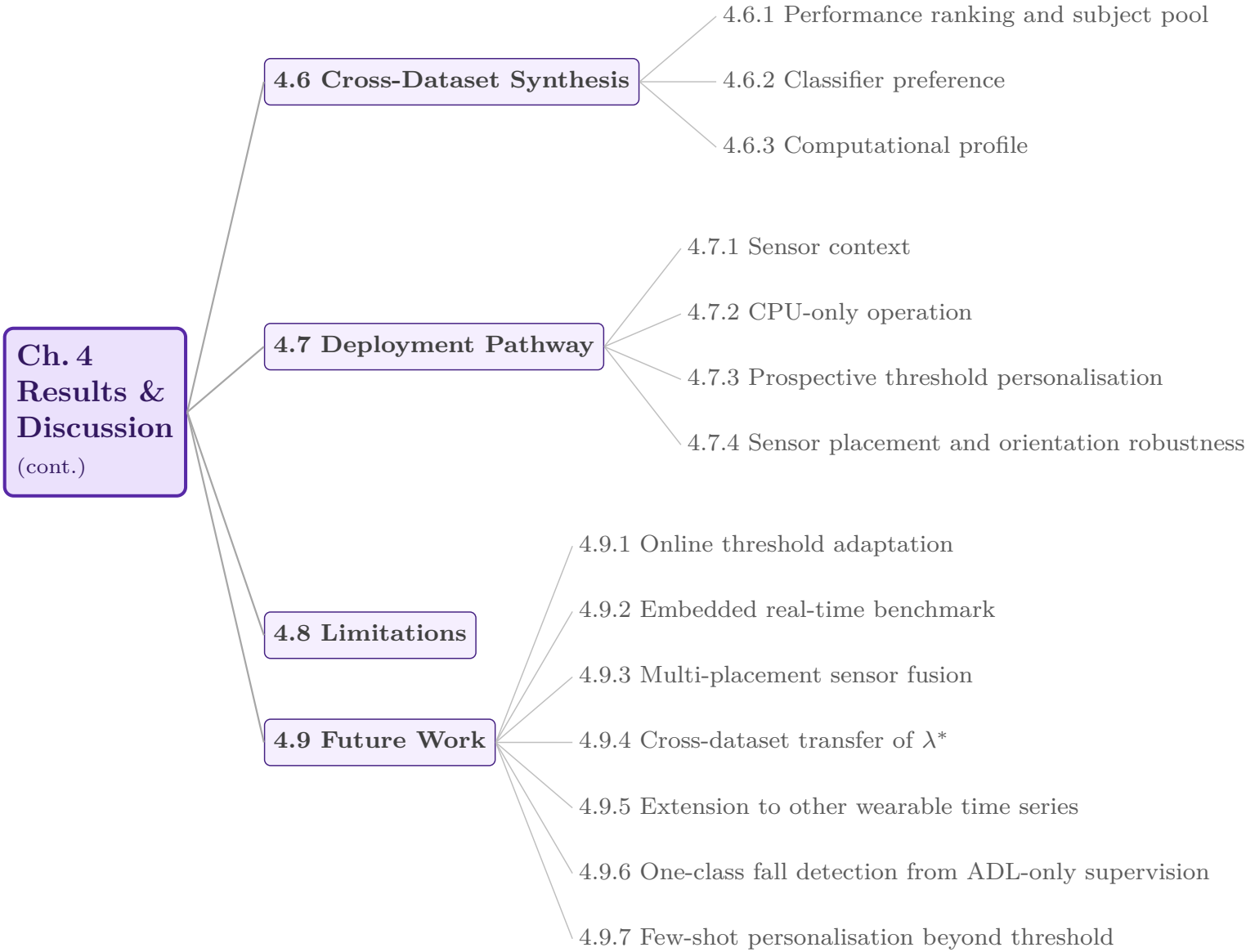


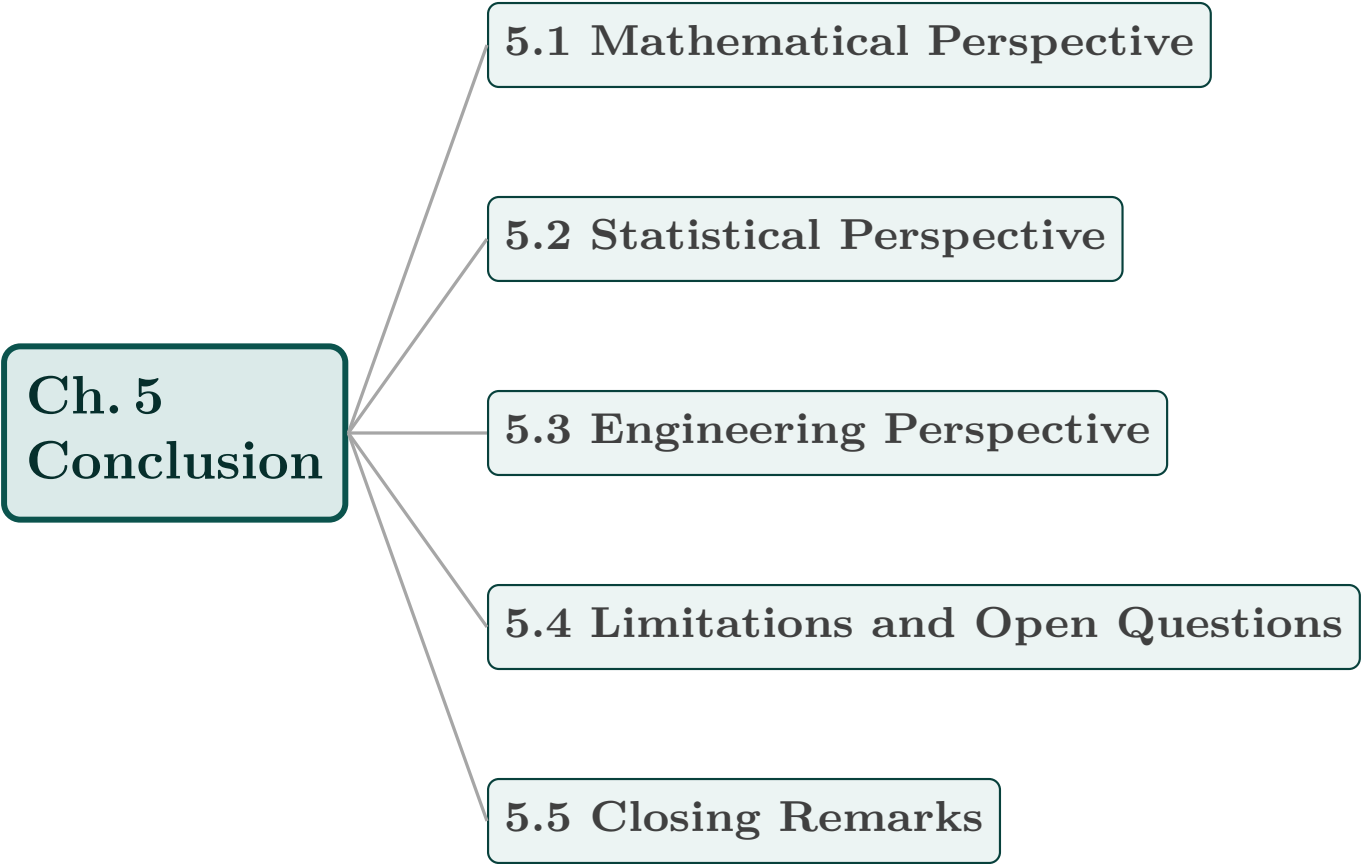




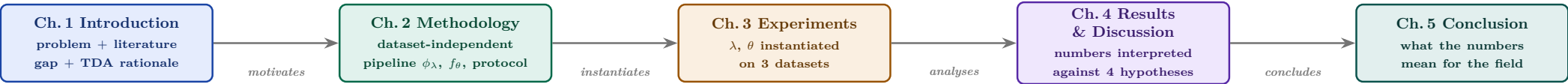








Each chapter’s tree above is not read in isolation: the leaves of one chapter are the reason the next chapter’s root exists. This is the macro-level dependency, corresponding to the vertical arrows a reader should keep in mind while moving from one tree to the next.

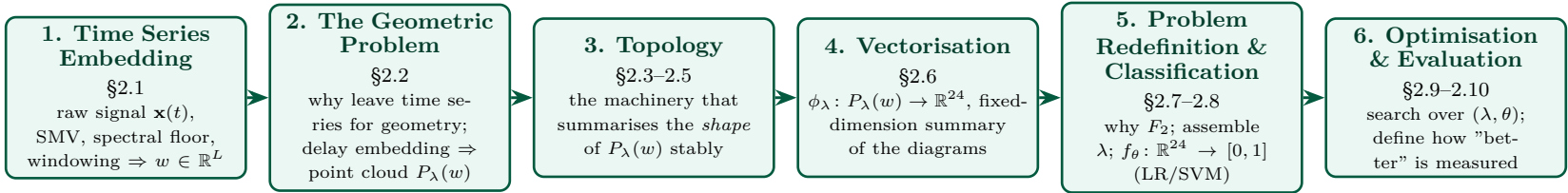


motivates: why a geometric, recall-oriented method is worth trying. instantiates: fixes λ, θ ranges and evaluation rules with concrete numbers. analyses: turns raw per-dataset numbers into evidence for/against each hypothesis. concludes: generalises the findings and states their limits.

Why this order, and not "topology first". A reader who has not seen the data has no reason to care about persistent homology. The chapters are therefore ordered so that *motivation always precedes machinery*: Chapter 1 establishes the problem and shows that existing methods leave a gap (Related Work → Limitations), then argues informally why a geometric representation should help. Chapter 2 only then commits to the machinery in the same order it is used — and that internal order is the subject of Part B, page 2.

Within Chapter 2, sections are *not* ordered by mathematical sophistication (that would put persistent homology, the hardest machinery, first). They are ordered by **when the pipeline needs the concept**: raw data first, geometry only once there is a reason to need it, optimisation last because it operates on everything defined before it.

This is no longer just an interpretive overlay: as of the current draft, the six stages below *are* the chapter’s actual section order (§2.1–§2.10), not a regrouping imposed after the fact.



Reading rule for stage 3. Topology only appears once stage 2 has already produced a point cloud in $\mathbb{R}^m$ — a purely geometric object with no reference to ”sensors” or ”falls” left in it. This is deliberate: it lets Chapter 2 argue that the representation map ϕ_λ is dataset- and experiment-independent, which is exactly what §2.9–2.10 need in order to be reused unchanged across all three datasets in Chapter 3.

Reading rule for stage 5. The hyperparameter tuple λ (§2.7.2) and the representation problem (§2.7.3) are stated here as a *recap*, not a first introduction: every component of λ and every constraint on ϕ_λ was already established in stages 1–4 at the point it first arose. Stage 5 only collects them into the formal classification problem, which is also why the F_2 justification (§2.7.1) now comes *before* the classifier is defined — the old draft defined the classifier’s decision rule before justifying the metric it depends on.

Everything in Chapter 2 is defined once, without numbers. The table below is the index a reader needs to find where each abstract object is later given concrete values (Chapter 3) or interpreted (Chapter 4).

Defined in Ch. 2 (Methodology)	Instantiated in Ch. 3 (Experiments)	Discussed in Ch. 4 (Results & Discussion)
§2.1.2 Spectral sufficiency ($f_s \geq 40$ Hz)	§3.1.1 Common target sampling rate (resampling to 50 Hz)	—
§2.2.1 Delay embedding $P_\lambda(w)$	§3.3.1 Persistence-Based Features (computed per dataset)	§4.2.2 Interpretation of configuration differences
§2.5 Persistent homology $\{D_k\}_{k=0,1}$	§3.3.1 Persistence-Based Features	§4.5 Ablation: H_0 vs. H_1 contribution
§2.6 Feature vector $\phi_\lambda \in \mathbb{R}^{24}$	§3.3 Feature Description (10+10+4 split)	§4.1 H1: F_2 outcomes attributed to ϕ_λ , not the classifier
§2.7.1 Performance metric selection (F_2)	§3.6 Results (headline scores)	§4.1.1 F_2 outcomes
§2.7.2 Hyperparameter tuple λ (recap)	§3.4 Bayesian HPO: search space, $T=200$ trials/dataset	§4.2 H2: distinct optimal λ^* per dataset
§2.8 Classifiers LR / SVM	§3.5 Classifiers and Training Setup	§4.3 H3: representation dominates classifier choice
§2.9 Bayesian HPO framework (TPE)	§3.4 Bayesian Hyperparameter Optimisation (engine run)	§4.2.3 Sensitivity of F_2 to λ
§2.10.1 Performance metrics (recap pointer to §2.7.1)	§3.6 Results, headline scores	§4.1.1 F_2 outcomes
§2.10.2 Evaluation protocols (LOSO, general/personal, threshold sweep)	§3.6 Results, computed under these protocols	§4.4 H4: threshold personalisation
§2.10.3 Statistical inference (t -CI, bootstrap, permutation test, generic)	§3.7 Statistical Inference: $R=15$, $B=10,000$, $\alpha=0.05$ instantiated	§4.1–4.4 all hypothesis discussions rely on these tests

Removed: the in-thesis dependency figure that used to open Chapter 2 (ch3.tex, fig:methodology-dependencies) was a 6-row need/solution/theorem diagram that, once its stage labels were updated to the current section numbers, no longer fit legibly on a portrait page — six stacked rows need more vertical room than a 145mm-wide column can give without shrinking the text below body size. Simplifying it (folding the theorem column into each solution box) only reduced clutter, not the underlying row count, so it was removed outright rather than moved to a landscape page. The chapter’s own intro paragraph and the pipeline-overview figure (now Figure 2.1) already carry the same information at a readable size; the six-stage concept-flow diagram on the previous page of this map is the closest surviving equivalent of what the removed figure showed.

Resolved: the statistical-testing machinery (t -confidence intervals, bootstrap, permutation tests, including the theorem environment for the t -CI) now has an abstract, dataset-independent home at §2.10.3 (Chapter 2). §3.7 (Experiments) was trimmed to a short paragraph instantiating the concrete settings ($R=15$, $B=10,000$, $\alpha=0.05$) and the two dataset-specific applications (personalisation distribution, cross-applied hyperparameters); the redundant “Methodological note on headline maxima” subsection was removed from Experiments since the same argument now lives as Remark 2.12 in §2.10.3. Confirmed by recompiling tez.tex: 155 pages, no undefined references, no duplicate labels.